

Assignment-1

Submission Instructions

Upload both *.ipynb with all relevant outputs (loss vs iteration curves, calculated SNR values, visualization of principal components and autoencoders weights) embedded and corresponding *.html files. Submit a screenshot in jpg format showing usage of BITS provided infrastructure for this assignment.

Marks will be deducted for inadequate training resulting and higher or unexpected results in all tasks! Your results should validate the well-established characteristics of different autoencoder architectures for the provided datasets.

Problem Statement

This assignment is about representation learning using variants of autoencoders.

Use (1) the CIFAR10 dataset provided in Keras, after conversion to gray-level images, and (2) the MNIST handwritten gray-scale database! Rescale the datasets so that both contain 28x28 images, and normalize the intensity levels of datasets between 50 and 200. Use randomly selected 70% of the dataset as training set, 20% as the validation, and 10% as test data.

Task 1: Perform standard PCA on the mean-centered training data for each dataset and identify principal components associated with top 30 eigenvalues (for the two datasets) and retain them. Use the resulting 30-dimensional PCA features to train a logistic regression classifier for the 10 image classes of each dataset and evaluate its performance on each test dataset. Plot the ROC curves for the test dataset for all 10 classes.

Repeat the experiment using randomized PCA with top 30 components and compare its results with standard PCA. Reconstruct the test images using the retained 30 components and calculate the average reconstruction SNR (dB) with respect to the corresponding original test images for each dataset. **[3 marks]**

Task 2: Train a single-layer linear autoencoder with a 30-dimensional bottleneck using the [50,200]-normalized data after mean-centering using the training-set mean, consistent with the PCA preprocessing in Task 1. Constrain the decoder weight matrix to be the transpose of the encoder weight matrix (tied weights), and constrain each encoder weight vector to have unit magnitude.

Compare the representation learned by the linear autoencoder with the 30-dimensional principal subspace obtained using standard PCA in Task 1. Display the top 30 PCA eigenvectors and the corresponding autoencoder weight vectors as grayscale images for qualitative comparison. Quantitatively compare the subspaces spanned by the PCA eigenvectors and the autoencoder weight vectors using an appropriate subspace-comparison metric. Report the metric obtained and justify its significance in determining the degree of alignment or similarity between the two learned subspaces.

Also train a logistic regression classifier using the 30-dimensional autoencoder features and compare the classification results with those obtained using the standard PCA features in Task 1 for each dataset. **[3 marks]**

Task 3: Investigate the effect of nonlinearity, network depth, and convolutional architecture on image reconstruction for both datasets.

Using a 30-dimensional latent representation, train a nonlinear shallow autoencoder with single hidden layer and a symmetric deep dense autoencoder with three hidden layers, including the 30-dimensional bottleneck layer. Use suitable nonlinear activation functions in the hidden layers and a linear activation function in the final reconstruction layer. Also design and train an appropriate deep convolutional autoencoder using the same 30-dimensional latent representation.

For each architecture, report the network architecture, number and size of hidden layers (or convolutional filters, as applicable), total number of trainable parameters, training and validation reconstruction loss, and average training and test reconstruction SNR (dB). Compare the reconstruction performance of

PCA-30 vs Shallow Nonlinear AE-30 vs Deep Dense AE-30 vs Deep CNN-AE-30.

Analyze the results to determine (i) whether introducing a nonlinear encoding-decoding mapping improves reconstruction relative to the linear PCA representation at the same representation dimensionality, (ii) whether additional successive nonlinear transformations through increased depth improve reconstruction while keeping the bottleneck dimensionality fixed, and (iii) whether the convolutional autoencoder provides additional reconstruction benefits through its ability to exploit local spatial relationships in image data. **[4 marks]**